

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 02 · Working with Vectors

With Answers

Questions with full worked solutions

- 66 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 02

Working with Vectors

Your classified route

30

QUESTIONS

323

MARKS

4

FAMILIES

11

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 02 | FAMILY 01

Resolve vectors and form resultants

WHAT THIS FAMILY ASKS

Convert magnitudes/directions into i-j components, add vectors and recover magnitude or direction.

VARIANT ROUTE

V01 Bearing or angle to components

V02 Decompose into prescribed parallel directions

V03 Resultant carried into dynamics

V04 Unknown components from a required resultant direction

V05 Non-component force triangle

10 classified questions

85 total marks

October 2019 Q7 — complete original question

7. Two forces, $\mathbf{F}$ and $\mathbf{G}$, act on a particle. The force $\mathbf{F}$ has magnitude 4 N and acts in a direction with a bearing of 120° and the force $\mathbf{G}$ has magnitude 6 N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{P}$
- (ii) the direction of $\mathbf{P}$, giving your answer as a bearing to the nearest degree.

(7)

DR ESLAM AHMED

October 2019 Q7 — detailed solution

Bearing to components

ALL
7 marks**Resolve 2F**

Since **F** has magnitude 4 N on a bearing of 120° , **2F** has magnitude 8 N on the same bearing.

Write the components

$2\mathbf{F} = 8 \sin 120^\circ \mathbf{i} + 8 \cos 120^\circ \mathbf{j} = 4\sqrt{3}\mathbf{i} - 4\mathbf{j}$. Also $\mathbf{G} = 6\mathbf{j}$.

Find P

$\mathbf{P} = 2\mathbf{F} + \mathbf{G} = 4\sqrt{3}\mathbf{i} + 2\mathbf{j}$.

Find the magnitude

$|\mathbf{P}| = \sqrt{(4\sqrt{3})^2 + 2^2} = \sqrt{48 + 4} = \sqrt{52} \text{ N}$.

Find the bearing angle

The vector is in the north-east quadrant. The angle east of north satisfies $\tan \theta = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$.

State the bearing

$\theta = 73.9^\circ$ approximately, so the bearing is 074° to the nearest degree.

Final answer (i) $|\mathbf{P}| = \sqrt{52} \text{ N} \approx 7.21 \text{ N}$. (ii) bearing = 074° .

January 2021 Q5 — complete original question

5. A particle is acted upon by two forces $\mathbf{F}$ and $\mathbf{G}$. The force $\mathbf{F}$ has magnitude 8 N and acts in a direction with a bearing of 240° . The force $\mathbf{G}$ has magnitude 10 N and acts due South.

Given that $\mathbf{R} = \mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{R}$,
- (ii) the direction of $\mathbf{R}$, giving your answer as a bearing to the nearest degree.

(7)

DR ESLAM AHMED

January 2021 Q5 — detailed solution

Bearing to components

ALL
7 marks**Resolve $\mathbf{F}$ into components**

A bearing of 240° is 60° west of south. Hence $\mathbf{F} = (-8 \sin 60^\circ)\mathbf{i} + (-8 \cos 60^\circ)\mathbf{j} = -4\sqrt{3}\mathbf{i} - 4\mathbf{j}$.

Add $\mathbf{G}$

The force $\mathbf{G}$ is due south with magnitude 10, so $\mathbf{G} = -10\mathbf{j}$. Therefore $\mathbf{R} = \mathbf{F} + \mathbf{G} = -4\sqrt{3}\mathbf{i} - 14\mathbf{j}$.

Find the magnitude

$$|\mathbf{R}| = \sqrt{(-4\sqrt{3})^2 + (-14)^2} = \sqrt{48 + 196} = \sqrt{244} \text{ N.}$$

Find the bearing angle

The resultant points south-west. The angle west of south satisfies $\tan \theta = \frac{4\sqrt{3}}{14} = \frac{2\sqrt{3}}{7}$.

State the bearing

$\theta = 26.3^\circ$. Bearing $= 180^\circ + 26.3^\circ = 206.3^\circ$, so the bearing is 206° to the nearest degree.

Final answer (i) $|\mathbf{R}| = \sqrt{244} \text{ N} \approx 15.6 \text{ N}$. (ii) bearing $= 206^\circ$.

Summer 2024 Q2 — complete original question

2. Two forces, $\mathbf{P}$ and $\mathbf{Q}$, act on a particle.

- $\mathbf{P}$ has magnitude 10 N and acts due west
- $\mathbf{Q}$ has magnitude 8 N and acts on a bearing of 330°

Given that $\mathbf{F} = \mathbf{P} + \mathbf{Q}$, find the magnitude of $\mathbf{F}$.

(4)

DR ESLAM AHMED

Summer 2024 Q2 — detailed solution

Bearing to components

ALL
4 marks**Resolve the two forces**

Take east as $\mathbf{i}$ and north as $\mathbf{j}$. The force due west is $-10\mathbf{i}$. A bearing of 330° is 30° west of north, so the 8 N force has components $-8 \sin 30^\circ \mathbf{i} + 8 \cos 30^\circ \mathbf{j}$.

Add the components

$$\mathbf{F} = (-10 - 4)\mathbf{i} + 4\sqrt{3}\mathbf{j} = -14\mathbf{i} + 4\sqrt{3}\mathbf{j}.$$

Find the magnitude

$$|\mathbf{F}| = \sqrt{(-14)^2 + (4\sqrt{3})^2} = \sqrt{196 + 48} = \sqrt{244} = 2\sqrt{61} \text{ N}.$$

Give a decimal value

$2\sqrt{61} = 15.620\dots$, so the magnitude is about 16 N to 2 significant figures.

Final answer $|\mathbf{F}| = 2\sqrt{61} \text{ N} \approx 15.6 \text{ N}.$

DR ESAM AHMED

Summer 2026 Q6 — complete original question

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

(a) Find the magnitude of $\mathbf{R}$.

(2)

(b) Find, in degrees, the angle between the direction of $\mathbf{R}$ and the direction of the vector $(2\mathbf{i} + \mathbf{j})$. Show your method clearly.

(3)

Given that $\mathbf{F}_1$ is parallel to the vector $(-\mathbf{i} + 2\mathbf{j})$ and that $\mathbf{F}_2$ is parallel to the vector $(2\mathbf{i} - 3\mathbf{j})$,

(c) find $\mathbf{F}_1$ and $\mathbf{F}_2$

(4)

DR ESLAM AHMED

Summer 2026 Q6 — detailed solution

Decompose into prescribed parallel directions

ALL
9 marks**Part (a): magnitude of the resultant**For $\mathbf{R} = -5\mathbf{i} + 6\mathbf{j}$, $|\mathbf{R}| = \sqrt{(-5)^2 + 6^2} = \sqrt{61} \text{ N}$.**Part (b): use the dot product**With $\mathbf{q} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{R} \cdot \mathbf{q} = -4$, $|\mathbf{q}| = \sqrt{5}$, so $\cos \theta = -4/\sqrt{305}$ and $\theta = 103^\circ$ to the nearest degree.**Part (c): introduce scalar multiples**Write $\mathbf{F}_1 = a(-\mathbf{i} + 2\mathbf{j})$ and $\mathbf{F}_2 = b(2\mathbf{i} - 3\mathbf{j})$. Equating their sum to $\mathbf{R}$ gives $-a + 2b = -5$ and $2a - 3b = 6$.**Solve and reconstruct the vectors**The equations give $a = -3$ and $b = -4$. Hence $\mathbf{F}_1 = 3\mathbf{i} - 6\mathbf{j}$ and $\mathbf{F}_2 = -8\mathbf{i} + 12\mathbf{j}$ newtons; their sum checks as $-5\mathbf{i} + 6\mathbf{j}$.**Final answer** (a) $\sqrt{61} \text{ N}$. (b) 103° . (c) $\mathbf{F}_1 = (3\mathbf{i} - 6\mathbf{j}) \text{ N}$, $\mathbf{F}_2 = (-8\mathbf{i} + 12\mathbf{j}) \text{ N}$.

DR ESLAM AHMED

Summer 2021 Q3 — complete original question

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

Given that P rests in equilibrium on a smooth horizontal surface under the action of these three forces,

- (a) find the size of the angle between the direction of $\mathbf{F}_3$ and the direction of $-\mathbf{j}$. (4)

The force $\mathbf{F}_3$ is now removed and replaced by the force $\mathbf{F}_4$ given by $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})\text{N}$, where λ is a positive constant.

When the three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_4$ act on P , the acceleration of P has magnitude 3.25 m s^{-2}

- (b) Find the value of λ . (5)

Summer 2021 Q3 — detailed solution

Resultant linked to dynamics

ALL
9 marks**Find F3 from equilibrium**Equilibrium gives $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$. Now $\mathbf{F}_1 + \mathbf{F}_2 = (5\mathbf{i} + 2\mathbf{j}) + (-3\mathbf{i} + \mathbf{j}) = 2\mathbf{i} + 3\mathbf{j}$.**Write F3**So $\mathbf{F}_3 = -2\mathbf{i} - 3\mathbf{j}$.**Find the angle with -j**The vector $-\mathbf{j}$ points vertically downward. For $\mathbf{F}_3 = -2\mathbf{i} - 3\mathbf{j}$, the angle from $-\mathbf{j}$ satisfies $\tan \theta = \frac{2}{3}$.**State the angle** $\theta = 33.7^\circ$, so the angle is about 34° .**Find the new resultant force**With $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})$, the resultant is $(2 + \lambda)\mathbf{i} + (3 + 3\lambda)\mathbf{j}$.**Use the acceleration magnitude**The mass is 4 kg and acceleration magnitude is 3.25, so the resultant force magnitude is $4(3.25) = 13$ N.**Form the equation**

$$(2 + \lambda)^2 + (3 + 3\lambda)^2 = 13^2.$$

Solve for lambdaExpanding gives $10\lambda^2 + 22\lambda - 156 = 0$. The roots are 3 and -5.2 , and since λ is positive, $\lambda = 3$.**Final answer** (a) 34° approximately. (b) $\lambda = 3$.

Summer 2023 Q2 — complete original question

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1$ N, $\mathbf{F}_2$ N and $\mathbf{F}_3$ N where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

Given that P remains in equilibrium,

- (a) find $\mathbf{F}_3$ in terms of c , $\mathbf{i}$ and $\mathbf{j}$. (2)

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

- (b) find the size of the angle between the direction of $\mathbf{i}$ and the direction of the resultant force acting on P . (4)

The mass of P is m kg.

Given that the magnitude of the acceleration of P is 8.5 ms^{-2}

- (c) find the value of m . (4)

Summer 2023 Q2 — detailed solution

Resultant linked to dynamics

ALL
10 marks**Use equilibrium**For equilibrium, $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$. Hence $\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2)$.**Find $\mathbf{F}_3$** $\mathbf{F}_1 + \mathbf{F}_2 = (3c - 14)\mathbf{i} + (4c + 7)\mathbf{j}$, so $\mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}$.**Find the resultant when $c = 2$** When $c = 2$, $\mathbf{F}_1 = 6\mathbf{i} + 8\mathbf{j}$ and $\mathbf{F}_2 = -14\mathbf{i} + 7\mathbf{j}$. Therefore the resultant is $-8\mathbf{i} + 15\mathbf{j}$.**Find the angle**The resultant is in the second quadrant. Its angle from the positive $\mathbf{i}$ direction is $180^\circ - \tan^{-1}\left(\frac{15}{8}\right) = 118.1^\circ$, so the angle is about 118° .**Find the magnitude of the resultant** $|\mathbf{R}| = \sqrt{(-8)^2 + 15^2} = 17 \text{ N}$.**Use $\mathbf{F} = m\mathbf{a}$** The acceleration magnitude is 8.5 m s^{-2} , so $17 = 8.5m$. Hence $m = 2$.**Final answer** (a) $\mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}$. (b) 118° approximately. (c) $m = 2$.

October 2025 Q3 — complete original question

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg moves on a smooth horizontal surface under the action of two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (-2\mathbf{i} + 3\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (4\mathbf{i} + 2\mathbf{j})\text{N}$.

- (a) Find the acceleration of P . (3)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ms}^{-1}$

- (b) Find the speed of P when $t = 3$ seconds. (4)

An additional force, $\mathbf{F}_3 = (b\mathbf{i} + c\mathbf{j})\text{N}$, is applied to P .

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ is equal to $\lambda(\mathbf{i} + \mathbf{j})\text{N}$, where λ is a constant.

- (c) Show that $b - c = 3$ (3)

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ has magnitude $10\sqrt{2}\text{N}$.

- (d) Find the two possible $\mathbf{F}_3$ forces. (4)

October 2025 Q3 — detailed solution

Resultant linked to dynamics

ALL
14 marks**Part (a): Find the acceleration of P - Find resultant force**

$$\mathbf{F}_{\text{resultant}} = \mathbf{F}_1 + \mathbf{F}_2$$

$$\mathbf{F}_{\text{resultant}} = (-2\mathbf{i} + 3\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j}) = (-2 + 4)\mathbf{i} + (3 + 2)\mathbf{j} = 2\mathbf{i} + 5\mathbf{j} \text{ N}$$

Part (a): Find the acceleration of P - Apply Newton's Second Law

$$\mathbf{F} = m\mathbf{a}$$

$$2\mathbf{i} + 5\mathbf{j} = 2 \times \mathbf{a} \quad \mathbf{a} = \frac{2\mathbf{i} + 5\mathbf{j}}{2} \quad \mathbf{a} = \mathbf{i} + 2.5\mathbf{j} \text{ m s}^{-2}$$

$$\text{Or: } \mathbf{a} = \mathbf{i} + \frac{5}{2}\mathbf{j} \text{ m s}^{-2}$$

Part (b): Find the speed of P when t = 3 seconds - Find velocity at t = 3

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

Given:

$$\mathbf{u} = 3\mathbf{i} - 4\mathbf{j} \text{ m s}^{-1}; \mathbf{a} = \mathbf{i} + 2.5\mathbf{j} \text{ m s}^{-2}; t = 3 \text{ seconds}$$

$$\mathbf{v} = (3\mathbf{i} - 4\mathbf{j}) + (\mathbf{i} + 2.5\mathbf{j}) \times 3 = 3\mathbf{i} - 4\mathbf{j} + 3\mathbf{i} + 7.5\mathbf{j} = 6\mathbf{i} + 3.5\mathbf{j} \text{ m s}^{-1}$$

Part (b): Find the speed of P when t = 3 seconds - Find the speed (magnitude of velocity)

$$|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$$

$$\text{Speed} = \sqrt{6^2 + 3.5^2} = \sqrt{36 + 12.25} = \sqrt{48.25} = 6.946... = 6.95 \text{ m s}^{-1} \text{ (3 s.f.)}$$

Part (c): Show that $\mathbf{b} - \mathbf{c} = 3$ - Find resultant of $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$

$$\mathbf{F}_{\text{resultant}} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = (-2\mathbf{i} + 3\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j}) + (b\mathbf{i} + c\mathbf{j}) = (-2 + 4 + b)\mathbf{i} + (3 + 2 + c)\mathbf{j} = (2 + b)\mathbf{i} + (5 + c)\mathbf{j}$$

Part (c): Show that $\mathbf{b} - \mathbf{c} = 3$ - Apply the conditionGiven that resultant = $\lambda(\mathbf{i} + \mathbf{j})$, we can write:

$$(2 + b)\mathbf{i} + (5 + c)\mathbf{j} = \lambda\mathbf{i} + \lambda\mathbf{j}$$

Part (c): Show that $\mathbf{b} - \mathbf{c} = 3$ - Compare coefficientsFor this to be in the form $\lambda(\mathbf{i} + \mathbf{j})$, the coefficients must be equal:

$$2 + b = \lambda \quad 5 + c = \lambda$$

$$\text{Therefore: } 2 + b = 5 + c \quad b = 5 + c - 2 \quad b = 3 + c \quad b - c = 3$$

Part (d): Find the two possible $\mathbf{F}_3$ forces - Use the magnitude condition

$$\text{Resultant} = (2 + b)\mathbf{i} + (5 + c)\mathbf{j}$$

$$\text{Since resultant} = \lambda(\mathbf{i} + \mathbf{j}), \text{ we have } 2 + b = 5 + c = \lambda$$

$$|\mathbf{F}_{\text{resultant}}| = \lambda\sqrt{1^2 + 1^2} = \lambda\sqrt{2}$$

$$\text{Given: } |\mathbf{F}_{\text{resultant}}| = 10\sqrt{2}$$

$$\lambda\sqrt{2} = 10\sqrt{2} \quad \lambda = 10 \quad \text{or} \quad \lambda = -10$$

(The negative value comes from considering the opposite direction)

Part (d): Find the two possible $\mathbf{F}_3$ forces - Find b and c for each value of λ

$$\text{Case 1: } \lambda = 10$$

$$2 + b = 10 \Rightarrow b = 8 \quad 5 + c = 10 \Rightarrow c = 5$$

$$\text{Check: } b - c = 8 - 5 = 3 \quad \checkmark$$

Therefore: $\mathbf{F}_3 = 8\mathbf{i} + 5\mathbf{j}$ N

Case 2: $\lambda = -10$

$$2 + b = -10 \Rightarrow b = -12 \quad 5 + c = -10 \Rightarrow c = -15$$

Check: $b - c = -12 - (-15) = 3 \checkmark$

Therefore: $\mathbf{F}_3 = -12\mathbf{i} - 15\mathbf{j}$ N

Final answer (a) $\mathbf{a} = \mathbf{i} + 2.5\mathbf{j} \text{ m s}^{-2}$. (b) speed = 6.95 m s^{-1} . (c) $b - c = 3$. (d) $\mathbf{F}_3 = (8\mathbf{i} + 5\mathbf{j}) \text{ N}$ or $(-12\mathbf{i} - 15\mathbf{j}) \text{ N}$.

DR ESLAM AHMED

October 2022 Q6 — complete original question

6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j}) \text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j}) \text{ N}$, where p and q are constants, are applied to A .

Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$ (4)

Given that $p = 5$

(b) find the speed of A at time $t = 4$ seconds. (5)

DR ESLAM AHMED

October 2022 Q6 — detailed solution

Required resultant direction

ALL
9 marks**Find the resultant force**

$$\mathbf{F} = (-3\mathbf{i} + 2\mathbf{j}) + (p\mathbf{i} + q\mathbf{j}) = (p - 3)\mathbf{i} + (q + 2)\mathbf{j}.$$

Use the direction of motion

Since the particle starts from rest, its motion is in the direction of the resultant force. This direction is $\mathbf{i} - 2\mathbf{j}$, so $\frac{q + 2}{p - 3} = -2$.

Show the relation

$$q + 2 = -2(p - 3) = -2p + 6, \text{ hence } 2p + q - 4 = 0.$$

Find q when p = 5

Substitute $p = 5$ into $2p + q - 4 = 0$: $10 + q - 4 = 0$, so $q = -6$.

Find the resultant force

The resultant force is $(5 - 3)\mathbf{i} + (-6 + 2)\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$ N.

Find acceleration

The mass is 0.5 kg, so $\mathbf{a} = \frac{2\mathbf{i} - 4\mathbf{j}}{0.5} = 4\mathbf{i} - 8\mathbf{j}$.

Find velocity at t = 4

Starting from rest, $\mathbf{v} = \mathbf{a}t = (4\mathbf{i} - 8\mathbf{j})4 = 16\mathbf{i} - 32\mathbf{j}$.

Find the speed

Speed = $\sqrt{16^2 + (-32)^2} = \sqrt{1280} = 16\sqrt{5} \text{ m s}^{-1}$, about 36 m s^{-1} .

Final answer (a) $2p + q - 4 = 0$. (b) speed = $16\sqrt{5} \text{ m s}^{-1} \approx 36 \text{ m s}^{-1}$.

January 2025 Q1 — complete original question

Clarification Notice

There is an error on page 2, Question 1b.

Please amend your question paper as follows:

The question currently reads:

- The resultant of the four forces acts in **the direction of** the vector $(7\mathbf{i} + 2\mathbf{j})$.

It should read:

- The resultant of the four forces acts in **a direction that is parallel to** the vector $(7\mathbf{i} + 2\mathbf{j})$.

1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{N}$$

- (a) Find the magnitude of the acceleration of the particle.

(4)

A fourth force, $\mathbf{F}_4 = (p\mathbf{i} + p\mathbf{j})\text{N}$, where p is a constant, is added.

The resultant of the four forces acts in the direction of the vector $(7\mathbf{i} + 2\mathbf{j})$.

- (b) Find the value of p .

(4)

January 2025 Q1 — detailed solution

Required resultant direction

ALL
8 marks**Find the resultant of the first three forces**

$$\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = (6\mathbf{i} + 8\mathbf{j}) + (-16\mathbf{i} + 2\mathbf{j}) + (-2\mathbf{i} + 8\mathbf{j}) = -12\mathbf{i} + 18\mathbf{j}.$$

Apply Newtons second law

The mass is 2.5 kg, so $-12\mathbf{i} + 18\mathbf{j} = 2.5\mathbf{a}$. Hence $\mathbf{a} = -4.8\mathbf{i} + 7.2\mathbf{j} \text{ m s}^{-2}$.

Find the magnitude of the acceleration

$|\mathbf{a}| = \sqrt{(-4.8)^2 + 7.2^2} = 8.653\dots$, so the magnitude is 8.7 m s^{-2} to 2 significant figures. Exactly,

$$|\mathbf{a}| = \frac{12\sqrt{13}}{5}.$$

Add the fourth force

With $\mathbf{F}_4 = p\mathbf{i} + p\mathbf{j}$, the new resultant is $(p - 12)\mathbf{i} + (p + 18)\mathbf{j}$.

Use the given direction

The resultant acts in the direction $7\mathbf{i} + 2\mathbf{j}$, so $\frac{p - 12}{p + 18} = \frac{7}{2}$.

Solve for p

$2(p - 12) = 7(p + 18)$, so $2p - 24 = 7p + 126$. Therefore $-150 = 5p$ and $p = -30$.

Final answer (a) $|\mathbf{a}| = \frac{12\sqrt{13}}{5} \text{ m s}^{-2} \approx 8.7 \text{ m s}^{-2}$. (b) $p = -30$.

January 2024 Q4 — complete original question

4.

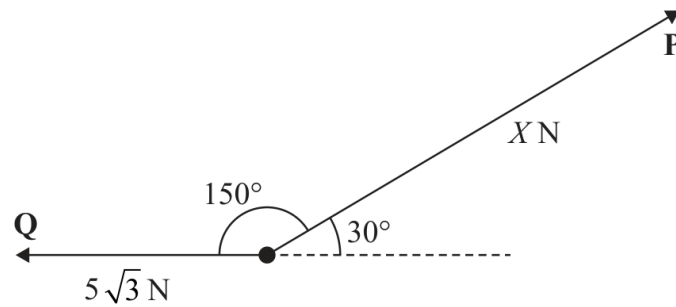


Figure 3

Figure 3 shows two horizontal forces **P** and **Q** acting on a particle.

The angle between the direction of **P** and the direction of **Q** is 150°

Force **P** has magnitude X newtons.

Force **Q** has magnitude $5\sqrt{3}$ N.

The resultant of **P** and **Q** has magnitude $\sqrt{129}$ N.

Find

- (i) the value of X .
- (ii) the angle between **Q** and the resultant, giving your answer to the nearest degree.

(8)

January 2024 Q4 — detailed solution

Force-triangle geometry

ALL
8 marks**Use the included angle in the force triangle**

The two given forces have angle 150° between them, so the triangle used for adding them has the supplementary angle 30° between the side of length X and the side of length $5\sqrt{3}$.

Use the cosine rule to find X

The resultant has magnitude $\sqrt{129}$. Therefore $129 = X^2 + (5\sqrt{3})^2 - 2X(5\sqrt{3})\cos 30^\circ$.

Solve for X

Since $(5\sqrt{3})^2 = 75$ and $2(5\sqrt{3})\cos 30^\circ = 15$, the equation becomes $129 = X^2 + 75 - 15X$. Thus $X^2 - 15X - 54 = 0$, giving $X = 18$.

Find the angle with Q

Let β be the angle between Q and the resultant. Using the cosine rule in the same triangle, $\frac{18}{\sin \beta} = \frac{\sqrt{129}}{\sin 30^\circ}$.

Choose the correct angle

This gives an obtuse angle for β . Equivalently, $\cos \beta = \frac{(5\sqrt{3})^2 + 129 - 18^2}{2(5\sqrt{3})\sqrt{129}}$, so $\beta = 127.6\dots^\circ$.

Round the angle

To the nearest degree, the angle between Q and the resultant is 128° .

Final answer (i) $X = 18$ N. (ii) angle = 128° to the nearest degree.

TOPIC 02 | FAMILY 02

Angle, bearing and direction constraints

WHAT THIS FAMILY ASKS

Determine or impose the direction of a vector using quadrants, bearings, angles, parallelism or a dot product.

VARIANT ROUTE

V01 Parallel, opposite or same-direction condition

4 classified questions

41 total marks

January 2020 Q6 — complete original question

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.

(a) Find the exact value of the magnitude of $\mathbf{F}$.

(2)

(b) Find, in degrees, the size of the angle between the direction of $\mathbf{F}$ and the direction of the vector $(\mathbf{i} + \mathbf{j})$.

(4)

The resultant of the force $\mathbf{F}$ and the force $(-15\mathbf{i} + a\mathbf{j})\text{N}$, where a is a constant, is parallel to, but in the opposite direction to, the vector $(2\mathbf{i} - 3\mathbf{j})$.

(c) Find the value of a .

(5)

DR ESLAM AHMED

January 2020 Q6 — detailed solution

Parallel or prescribed direction

ALL
11 marks**Find the magnitude**

For $\mathbf{F} = 10\mathbf{i} + \mathbf{j}$, $|\mathbf{F}| = \sqrt{10^2 + 1^2} = \sqrt{101}$ N.

Find the direction angle of $\mathbf{F}$

The angle that $\mathbf{F}$ makes with the positive $\mathbf{i}$ direction is α , where $\tan \alpha = \frac{1}{10}$. Thus $\alpha = 5.71^\circ$ approximately.

Compare with $\mathbf{i}$ plus $\mathbf{j}$

The vector $\mathbf{i} + \mathbf{j}$ makes an angle of 45° with the positive $\mathbf{i}$ direction.

Find the angle between the vectors

So the required angle is $45^\circ - 5.71^\circ = 39.29^\circ$, approximately 39.3° .

Write the resultant

The resultant of $10\mathbf{i} + \mathbf{j}$ and $-15\mathbf{i} + a\mathbf{j}$ is $-5\mathbf{i} + (a + 1)\mathbf{j}$.

Use the parallel opposite direction

Parallel but opposite to $2\mathbf{i} - 3\mathbf{j}$ means the resultant is a positive multiple of $-2\mathbf{i} + 3\mathbf{j}$.

Solve for a

Let $-5\mathbf{i} + (a + 1)\mathbf{j} = \lambda(-2\mathbf{i} + 3\mathbf{j})$. From the $\mathbf{i}$ component, $-5 = -2\lambda$, so $\lambda = 2.5$. Then $a + 1 = 3(2.5) = 7.5$, hence $a = 6.5$.

Final answer (a) $|\mathbf{F}| = \sqrt{101}$ N. (b) 39.3° approximately. (c) $a = 6.5$.

January 2022 Q6 — complete original question

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 2 kg moves under the action of two forces, $(p\mathbf{i} + q\mathbf{j})\text{ N}$ and $(2q\mathbf{i} + p\mathbf{j})\text{ N}$, where p and q are constants.

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j})\text{ m s}^{-2}$

(a) find the value of p and the value of q .

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector $\mathbf{j}$.

(2)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

At $t = T$ seconds, P is moving in the direction of the vector $(11\mathbf{i} - 13\mathbf{j})$.

(c) Find the value of T .

(5)

DR ESAM AHMED

January 2022 Q6 — detailed solution

Parallel or prescribed direction

ALL
12 marks**Use $F = ma$**

The total force is $(p\mathbf{i} + q\mathbf{j}) + (2q\mathbf{i} + p\mathbf{j}) = (p + 2q)\mathbf{i} + (p + q)\mathbf{j}$. Since the mass is 2 kg and acceleration is $\mathbf{i} - \mathbf{j}$, the total force is $2\mathbf{i} - 2\mathbf{j}$.

Equate components

$p + 2q = 2$ and $p + q = -2$.

Find p and q

Subtract the equations to get $q = 4$. Then $p + 4 = -2$, so $p = -6$.

Find the angle with $\mathbf{j}$

The acceleration is $\mathbf{i} - \mathbf{j}$. Its dot product with $\mathbf{j}$ is -1 , so the angle with the positive $\mathbf{j}$ direction is obtuse. Since the components are equal in magnitude, the angle is 135° .

Write the velocity at time T

$\mathbf{v} = (3\mathbf{i} - 4\mathbf{j}) + T(\mathbf{i} - \mathbf{j}) = (3 + T)\mathbf{i} + (-4 - T)\mathbf{j}$.

Use the direction vector

The velocity is parallel to $11\mathbf{i} - 13\mathbf{j}$, so $\frac{3 + T}{11} = \frac{-4 - T}{-13}$.

Solve for T

$13(3 + T) = 11(4 + T)$, so $39 + 13T = 44 + 11T$. Hence $2T = 5$ and $T = 2.5$.

Final answer (a) $p = -6$, $q = 4$. (b) 135° . (c) $T = 2.5$.

January 2023 Q3 — complete original question

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

- (a) Find the speed of P at time $t = 2$ seconds. (3)

- (b) Find the size of the angle between the direction of $\mathbf{i}$ and the direction of motion of P at time $t = 2$ seconds. (3)

At time $t = T$ seconds, P is moving in the direction of vector $(2\mathbf{i} - 3\mathbf{j})$

- (c) Find the value of T (4)

DR ESLAM AHMED

January 2023 Q3 — detailed solution

Parallel or prescribed direction

ALL
10 marks**Find the velocity at $t = 2$**

$$\mathbf{v} = (14\mathbf{i} - 5\mathbf{j}) + 2(-4\mathbf{i} + \mathbf{j}) = 6\mathbf{i} - 3\mathbf{j}.$$

Find the speed

$$\text{The speed is } \sqrt{6^2 + (-3)^2} = \sqrt{45} = 3\sqrt{5} \text{ m s}^{-1}.$$

Find the angle with $\mathbf{i}$

The velocity $6\mathbf{i} - 3\mathbf{j}$ is below the positive $\mathbf{i}$ direction. The acute angle satisfies $\tan \theta = \frac{3}{6}$.

State the angle

$\theta = 26.6^\circ$, so the size of the angle is 27° to the nearest degree.

Write the velocity at time T

$$\mathbf{v}(T) = (14 - 4T)\mathbf{i} + (-5 + T)\mathbf{j}.$$

Use the given direction

The direction is parallel to $2\mathbf{i} - 3\mathbf{j}$, so the component ratio must match: $\frac{14 - 4T}{2} = \frac{-5 + T}{-3}$.

Solve for T

$$-3(14 - 4T) = 2(-5 + T), \text{ so } -42 + 12T = -10 + 2T. \text{ Hence } 10T = 32 \text{ and } T = 3.2.$$

Final answer (a) $3\sqrt{5} \text{ m s}^{-1} \approx 6.7 \text{ m s}^{-1}$. (b) 27° . (c) $T = 3.2$.

January 2026 Q3 — complete original question

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The resultant force acts in the direction of the vector $(4\mathbf{i} - 5\mathbf{j})$

- (a) Show that

$$2p - q = 19 \quad (4)$$

The mass of the particle is 0.5 kg.

Given that $p = 7$

- (b) find the acceleration of the particle.

(4)

January 2026 Q3 — detailed solution

Parallel or prescribed direction

ALL
8 marks**Add the three forces**

The resultant force is $(2p\mathbf{i} - 3p\mathbf{j}) + (q\mathbf{i} + \mathbf{j}) + (3\mathbf{i} - q\mathbf{j}) = (2p + q + 3)\mathbf{i} + (-3p + 1 - q)\mathbf{j}$.

Use the direction vector

The resultant acts in the direction $4\mathbf{i} - 5\mathbf{j}$, so the ratio of the components is the same: $\frac{-3p + 1 - q}{2p + q + 3} = \frac{-5}{4}$.

Show the required relation

Cross-multiplying gives $4(-3p + 1 - q) = -5(2p + q + 3)$. Therefore $-12p + 4 - 4q = -10p - 5q - 15$, so $2p - q = 19$.

Find q when p is 7

With $p = 7$, the relation gives $14 - q = 19$, hence $q = -5$.

Apply Newtons second law

Substitute $p = 7$ and $q = -5$ into the resultant: $\mathbf{F} = 12\mathbf{i} - 15\mathbf{j}$. Since the mass is 0.5 kg , $\mathbf{F} = m\mathbf{a}$ gives $12\mathbf{i} - 15\mathbf{j} = 0.5\mathbf{a}$.

Find the acceleration

Dividing by 0.5 gives $\mathbf{a} = 24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$.

Final answer (a) $2p - q = 19$. (b) $\mathbf{a} = 24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$.

TOPIC 02 | FAMILY 03

Position vectors and relative motion

WHAT THIS FAMILY ASKS

Model one or more moving particles with $r = r_0 + vt$ and use relative position to answer meeting, direction or distance questions.

VARIANT ROUTE

V01 Meeting or route intersection

V02 Relative bearing or distance at a time

V03 Infer a landmark or route point

V04 Timed position with distance/angle locus

15 classified questions

186 total marks

October 2023 Q6 — complete original question

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})\text{ km}$.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})\text{ km}$.

Given that P moves with constant velocity,

- (a) show that the velocity of P is $(10\mathbf{i} - 4\mathbf{j})\text{ km h}^{-1}$ (2)

At time t hours after 12:00, the position vector of P is $\mathbf{p}\text{ km}$.

- (b) Find an expression for $\mathbf{p}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t . (2)

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}\text{ km}$, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships P and Q are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30
 (ii) find the position vector of the point at which the collision would occur. (4)

At 12:30 Q changes speed and direction to avoid the collision.

Ship Q now travels due north with a constant speed of 15 km h^{-1}

Ship P maintains the same constant velocity throughout.

- (d) Find the exact distance between P and Q at 14:30 (7)

October 2023 Q6 — detailed solution

Meeting by matching components

ALL
15 marks**Find the velocity of P**

From 12:00 to 12:30, the displacement of P is $(20\mathbf{i} + 34\mathbf{j}) - (15\mathbf{i} + 36\mathbf{j}) = 5\mathbf{i} - 2\mathbf{j}$ km. The time is 0.5 hours, so $\mathbf{v}_P = \frac{5\mathbf{i} - 2\mathbf{j}}{0.5} = 10\mathbf{i} - 4\mathbf{j}$.

Write the position of P

At t hours after 12:00, $\mathbf{p} = (15\mathbf{i} + 36\mathbf{j}) + t(10\mathbf{i} - 4\mathbf{j}) = (15 + 10t)\mathbf{i} + (36 - 4t)\mathbf{j}$.

Verify the collision time

At 13:30, $t = 1.5$. Then $\mathbf{p} = (15 + 15)\mathbf{i} + (36 - 6)\mathbf{j} = 30\mathbf{i} + 30\mathbf{j}$. Also $\mathbf{q} = (42 - 8(1.5))\mathbf{i} + (9 + 14(1.5))\mathbf{j} = 30\mathbf{i} + 30\mathbf{j}$.

State the collision position

Since both position vectors are the same, the collision point has position vector $30\mathbf{i} + 30\mathbf{j}$ km.

Find P at 14:30

At 14:30, $t = 2.5$, so $\mathbf{p} = (15 + 25)\mathbf{i} + (36 - 10)\mathbf{j} = 40\mathbf{i} + 26\mathbf{j}$.

Find Q after changing course

At 12:30, $t = 0.5$, so $\mathbf{q} = 38\mathbf{i} + 16\mathbf{j}$. From 12:30 to 14:30 is 2 hours due north at 15 km h^{-1} , so the extra displacement is $30\mathbf{j}$. Thus Q is at $38\mathbf{i} + 46\mathbf{j}$.

Find the distance

The separation vector is $(40 - 38)\mathbf{i} + (26 - 46)\mathbf{j} = 2\mathbf{i} - 20\mathbf{j}$. The distance is $\sqrt{2^2 + 20^2} = \sqrt{404} = 2\sqrt{101}$ km.

Final answer (a) $\mathbf{v}_P = 10\mathbf{i} - 4\mathbf{j} \text{ km h}^{-1}$. (b) $\mathbf{p} = (15 + 10t)\mathbf{i} + (36 - 4t)\mathbf{j}$. (c)(ii) $30\mathbf{i} + 30\mathbf{j}$. (d) $2\sqrt{101}$ km.

October 2024 Q3 — complete original question

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin.]

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

At time t hours after 1 pm, the position vector of A is $\mathbf{r}_A$ km.

- (a) Show that $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$ (4)

The speed of A is $V \text{ m s}^{-1}$

- (b) Find the value of V . (2)

A ship B is moving with constant velocity $(20\mathbf{i} - 6\mathbf{j}) \text{ km h}^{-1}$

At 1 pm, the position vector of B is $(35\mathbf{i} + 51\mathbf{j})$ km.

At 2:30 pm, B passes through the point P .

- (c) Show that A also passes through P . (5)

October 2024 Q3 — detailed solution

Meeting by matching components

ALL
11 marks**Find the velocity of A**

From 1 pm to 3 pm is 2 hours. The displacement of A is $(55\mathbf{i} + 34\mathbf{j}) - (25\mathbf{i} + 10\mathbf{j}) = 30\mathbf{i} + 24\mathbf{j}$ km, so $\mathbf{v}_A = 15\mathbf{i} + 12\mathbf{j}$ km h⁻¹.

Write the position vector of A

At time t hours after 1 pm, $\mathbf{r}_A = (25\mathbf{i} + 10\mathbf{j}) + t(15\mathbf{i} + 12\mathbf{j}) = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$.

Convert the speed to metres per second

The speed of A is $\sqrt{15^2 + 12^2} = \sqrt{369}$ km h⁻¹. Converting to m s⁻¹, $V = \sqrt{369} \times \frac{1000}{3600} = \frac{5\sqrt{41}}{6}$ m s⁻¹.

Find the point P reached by B

At 2:30 pm, $t = 1.5$ hours after 1 pm. For B, $\mathbf{r}_B = (35\mathbf{i} + 51\mathbf{j}) + 1.5(20\mathbf{i} - 6\mathbf{j}) = 65\mathbf{i} + 42\mathbf{j}$.

Check whether A reaches P

Set $\mathbf{r}_A = 65\mathbf{i} + 42\mathbf{j}$. From the i-component, $25 + 15t = 65$, so $t = \frac{8}{3}$.

Confirm with the j-component

When $t = \frac{8}{3}$, $10 + 12t = 10 + 32 = 42$. Both components match, so A also passes through P.

Final answer (a) $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$. (b) $V = \frac{5\sqrt{41}}{6}$ m s⁻¹ ≈ 5.3 m s⁻¹. (c) A also passes through P.

October 2019 Q8 — complete original question

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (a) Find the velocity of B in the form $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$ (3)
- (b) Find expressions for $\mathbf{r}$ and $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

October 2019 Q8 — detailed solution

Relative bearing and distance

ALL
14 marks**Resolve B's velocity**

A bearing of 150° is 30° east of south. Therefore the east component is $40 \sin 150^\circ = 20$ and the north component is $40 \cos 150^\circ = -20\sqrt{3}$.

Write B's velocity

$$\mathbf{v}_B = 20\mathbf{i} - 20\sqrt{3}\mathbf{j} \text{ km h}^{-1}.$$

Write A's position

Boat A starts at $60\mathbf{i}$ and travels due west at 20 km h^{-1} , so $\mathbf{r} = 60\mathbf{i} - 20t\mathbf{i} = (60 - 20t)\mathbf{i}$.

Write B's position

Boat B starts at O, so $\mathbf{s} = t(20\mathbf{i} - 20\sqrt{3}\mathbf{j}) = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$.

Find the separation vector

$$\overrightarrow{AB} = \mathbf{s} - \mathbf{r} = (20t - (60 - 20t))\mathbf{i} - 20\sqrt{3}t\mathbf{j} = (40t - 60)\mathbf{i} - 20\sqrt{3}t\mathbf{j}.$$

Use the noon distance

At noon the distance between the boats is 60 km. Set the later distance equal to 60: $(40t - 60)^2 + (-20\sqrt{3}t)^2 = 60^2$.

Solve for t

Expanding gives $2800t^2 - 4800t = 0$, so $t(2800t - 4800) = 0$. The later time is $t = \frac{4800}{2800} = \frac{12}{7}$ hours.

Convert to clock time

$\frac{12}{7}$ hours is 1 hour and $42.857 \dots$ minutes after noon, which rounds to 1:43 pm, or 13:43.

Final answer (a) $\mathbf{v}_B = 20\mathbf{i} - 20\sqrt{3}\mathbf{j} \text{ km h}^{-1}$. (b) $\mathbf{r} = (60 - 20t)\mathbf{i}$, $\mathbf{s} = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$. (c) 13:43, or 1:43 pm.

January 2022 Q8 — complete original question

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ km h}^{-1}$

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j})$ km.

- (a) Find $\mathbf{r}$ in terms of t .

(2)

Another ship B moves with constant velocity $(15\mathbf{i} + 14\mathbf{j}) \text{ km h}^{-1}$

At time $t = 0$, B is at the point with position vector $(3\mathbf{i} - 5\mathbf{j})$ km.

- (b) Show that, at time t hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km}$$

(4)

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

- (d) find the bearing of ship B from ship A when the ships are closest.

(2)

(Total 14 marks)

January 2022 Q8 — detailed solution

Relative bearing and distance

ALL
14 marks**Write the position of A**

$$\mathbf{r} = (13\mathbf{i} + 5\mathbf{j}) + t(3\mathbf{i} - 10\mathbf{j}) = (13 + 3t)\mathbf{i} + (5 - 10t)\mathbf{j}.$$

Write the position of B

$$\mathbf{s} = (3\mathbf{i} - 5\mathbf{j}) + t(15\mathbf{i} + 14\mathbf{j}) = (3 + 15t)\mathbf{i} + (-5 + 14t)\mathbf{j}.$$

Find AB

$$\overrightarrow{AB} = \mathbf{s} - \mathbf{r} = [3 + 15t - (13 + 3t)]\mathbf{i} + [-5 + 14t - (5 - 10t)]\mathbf{j}.$$

Show the expression

This simplifies to $\overrightarrow{AB} = (12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}$ km.

Use distance squared

The square of the distance is $d^2 = (12t - 10)^2 + (24t - 10)^2$.

Expand and minimise

$$d^2 = 720t^2 - 720t + 200. \text{ This quadratic is minimum at } t = \frac{720}{2 \cdot 720} = \frac{1}{2}.$$

Find the shortest distance

At $t = \frac{1}{2}$, $\overrightarrow{AB} = (-4)\mathbf{i} + 2\mathbf{j}$, so the shortest distance is $\sqrt{(-4)^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$ km.

Find the bearing

At closest approach, B is west and north of A. The angle west of north satisfies $\tan \theta = \frac{4}{2} = 2$, so $\theta = 63.4^\circ$. Bearing = $360^\circ - 63.4^\circ = 296.6^\circ$, which is 297° .

Final answer (a) $\mathbf{r} = (13 + 3t)\mathbf{i} + (5 - 10t)\mathbf{j}$. (b) $\overrightarrow{AB} = (12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}$. (c) $2\sqrt{5}$ km $\approx$ 4.5 km. (d) 297° .

Summer 2022 Q8 — complete original question

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ m s}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the direction in which Q is travelling, giving your answer as a bearing. (2)

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j} \text{ m}$.

At time t seconds, the position vector of P is $\mathbf{p} \text{ m}$ and the position vector of Q is $\mathbf{q} \text{ m}$.

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (c) Find the bearing of P from Q when $t = 10$ (2)

- (d) Find the distance between P and Q when Q is north east of P (5)

- (e) Find the times when P and Q are 200 m apart. (3)

(Total 17 marks)

Summer 2022 Q8 — detailed solution

Relative bearing and distance

ALL
17 marks**Find the bearing of Q**

Q has velocity $20\mathbf{i} - 20\mathbf{j}$, so it travels south-east with equal east and south components. Its bearing is 135° .

Write P position

P starts at the origin and has velocity $15\mathbf{i}$, so $\mathbf{p} = 15t\mathbf{i}$.

Write Q position

Q starts at $200\mathbf{j}$ and has velocity $20\mathbf{i} - 20\mathbf{j}$, so $\mathbf{q} = 20t\mathbf{i} + (200 - 20t)\mathbf{j}$.

Show PQ

$$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p} = (20t - 15t)\mathbf{i} + (200 - 20t)\mathbf{j} = 5t\mathbf{i} + (200 - 20t)\mathbf{j}.$$

Find the bearing at $t = 10$

At $t = 10$, $\overrightarrow{PQ} = 50\mathbf{i}$. This means Q is east of P, so P is west of Q. The bearing of P from Q is 270° .

Use the north-east condition

Q is north-east of P when the $\mathbf{i}$ and $\mathbf{j}$ components of $\overrightarrow{PQ}$ are equal and positive. So $5t = 200 - 20t$.

Find the distance then

The equation gives $t = 8$. Then $\overrightarrow{PQ} = 40\mathbf{i} + 40\mathbf{j}$, so the distance is $\sqrt{40^2 + 40^2} = 40\sqrt{2}$ m.

Set the 200 m distance equation

For distance 200 m, $(5t)^2 + (200 - 20t)^2 = 200^2$.

Solve for t

This simplifies to $425t^2 - 8000t = 0$, so $t(425t - 8000) = 0$. Hence $t = 0$ or $t = \frac{8000}{425} = \frac{320}{17}$ s.

Final answer (a) 135° . (b) $\overrightarrow{PQ} = 5t\mathbf{i} + (200 - 20t)\mathbf{j}$. (c) 270° . (d) $40\sqrt{2}$ m. (e) $t = 0$ s or $\frac{320}{17}$ s.

October 2022 Q8 — complete original question

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of A .

(2)

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$ and B is at the point with position vector $(16\mathbf{i} + 6\mathbf{j})\text{ km}$.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of p and the value of q .

(7)

- (c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

(Total 16 marks)

October 2022 Q8 — detailed solution

Relative bearing and distance

ALL
16 marks**Find the speed of A**

$$|\mathbf{v}_A| = \sqrt{3^2 + 12^2} = \sqrt{153} = 3\sqrt{17} \text{ km h}^{-1}.$$

Write A position

$$\text{At time } t, \mathbf{a} = (-9\mathbf{i} + 6\mathbf{j}) + t(3\mathbf{i} + 12\mathbf{j}) = (-9 + 3t)\mathbf{i} + (6 + 12t)\mathbf{j}.$$

Write B position

$$\mathbf{b} = (16\mathbf{i} + 6\mathbf{j}) + t(p\mathbf{i} + q\mathbf{j}) = (16 + pt)\mathbf{i} + (6 + qt)\mathbf{j}.$$

Use AB vector

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = [25 + (p - 3)t]\mathbf{i} + [(q - 12)t]\mathbf{j}.$$

Compare components

$$\text{Given } \overrightarrow{AB} = (25 - 12t)\mathbf{i} - 9t\mathbf{j}, \text{ compare coefficients: } p - 3 = -12 \text{ and } q - 12 = -9.$$

Find p and q

$$\text{Therefore } p = -9 \text{ and } q = 3.$$

Find the time when the distance is 15

$$\text{Use } |\overrightarrow{AB}| = 15: (25 - 12t)^2 + (-9t)^2 = 15^2.$$

Solve the distance equation

$$\text{This gives } 225t^2 - 600t + 400 = 0, \text{ so } (3t - 4)^2 = 0 \text{ and } t = \frac{4}{3}.$$

Find BA at this time

$$\text{At } t = \frac{4}{3}, \overrightarrow{AB} = 9\mathbf{i} - 12\mathbf{j}. \text{ Therefore } \overrightarrow{BA} = -9\mathbf{i} + 12\mathbf{j}.$$

Convert to a bearing

A is north-west of B. The angle west of north satisfies $\tan \theta = \frac{9}{12}$, so $\theta = 36.9^\circ$. Bearing = $360^\circ - 36.9^\circ = 323^\circ$ to the nearest degree.

Final answer (a) $3\sqrt{17} \text{ km h}^{-1}$. (b) $p = -9, q = 3$. (c) bearing = 323° .

Summer 2023 Q8 — complete original question

8.

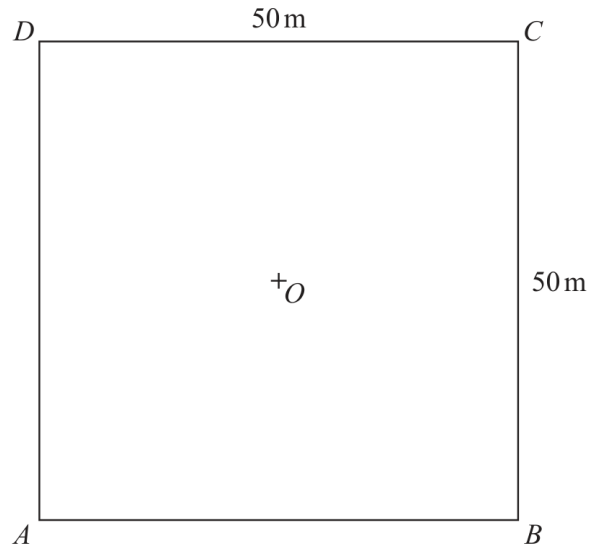


Figure 5

A square floor space $ABCD$, with centre O , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O .

A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})\text{ m}$
- At time $t = 11\text{ s}$, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})\text{ m}$
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

A second robot S is at the point C .

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

Summer 2023 Q8 — detailed solution

Relative bearing and distance

ALL
9 marks**Find the velocity of R**R moves from $-2\mathbf{i} + \mathbf{j}$ to $9\mathbf{i} + 23\mathbf{j}$ in 11 s. Its displacement is $11\mathbf{i} + 22\mathbf{j}$, so its velocity is $\mathbf{i} + 2\mathbf{j}$.**Write r**

$$\mathbf{r} = (-2\mathbf{i} + \mathbf{j}) + t(\mathbf{i} + 2\mathbf{j}) = (t - 2)\mathbf{i} + (2t + 1)\mathbf{j}.$$

Write sPoint C has position vector $25\mathbf{i} + 25\mathbf{j}$. Since S moves with velocity $-\mathbf{i} - \mathbf{j}$, $\mathbf{s} = (25 - t)\mathbf{i} + (25 - t)\mathbf{j}$.**Find SR vector**

$$\overrightarrow{SR} = \mathbf{r} - \mathbf{s} = [(t - 2) - (25 - t)]\mathbf{i} + [(2t + 1) - (25 - t)]\mathbf{j}.$$

Simplify SR

$$\text{Therefore } \overrightarrow{SR} = (2t - 27)\mathbf{i} + (3t - 24)\mathbf{j} \text{ m.}$$

Minimise the distance squared

$$\text{The distance squared is } d^2 = (2t - 27)^2 + (3t - 24)^2.$$

Expand the quadratic

$$d^2 = 4t^2 - 108t + 729 + 9t^2 - 144t + 576 = 13t^2 - 252t + 1305.$$

Find the minimum time

$$\text{The quadratic is minimum at } t = \frac{252}{2 \cdot 13} = \frac{126}{13} = 9.69 \dots \text{ s.}$$

Final answer (a) $\mathbf{r} = (t - 2)\mathbf{i} + (2t + 1)\mathbf{j}$. (b) $\mathbf{s} = (25 - t)\mathbf{i} + (25 - t)\mathbf{j}$. (c) $\overrightarrow{SR} = (2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}$. (d) $t = \frac{126}{13} \text{ s} \approx 9.7 \text{ s}$.

Summer 2024 Q7 — complete original question

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity.

- the velocity of A is 40 km h^{-1} due east
- the velocity of B is 20 km h^{-1} on a bearing of angle α ($0^\circ < \alpha < 90^\circ$), where $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B in km h^{-1} (2)

At noon

- the position vector of A is $20\mathbf{j}$ km
- the position vector of B is $(10\mathbf{i} + 5\mathbf{j})$ km

At time t hours after noon

- the position vector of A is $\mathbf{r}$ km, where $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of B is $\mathbf{s}$ km

- (b) Find an expression for $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (2)

- (c) Show that at time t hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km} \quad (2)$$

- (d) Show that the boats will never collide. (3)

- (e) Find the distance between the boats when the bearing of B from A is 225° (4)

Summer 2024 Q7 — detailed solution

Relative bearing and distance

ALL
13 marks**Resolve the velocity of B**

For a bearing angle α with $\tan \alpha = \frac{4}{3}$, use $\sin \alpha = \frac{4}{5}$ and $\cos \alpha = \frac{3}{5}$. Therefore $\mathbf{v}_B = 20 \sin \alpha \mathbf{i} + 20 \cos \alpha \mathbf{j} = 16\mathbf{i} + 12\mathbf{j}$.

Write the position of B

At noon, B is at $10\mathbf{i} + 5\mathbf{j}$. With constant velocity $16\mathbf{i} + 12\mathbf{j}$, $\mathbf{s} = (10 + 16t)\mathbf{i} + (5 + 12t)\mathbf{j}$.

Find AB vector

A has position vector $\mathbf{r} = 40t\mathbf{i} + 20\mathbf{j}$. Hence $\overrightarrow{AB} = \mathbf{s} - \mathbf{r} = (10 + 16t - 40t)\mathbf{i} + (5 + 12t - 20)\mathbf{j}$.

Show the printed expression

This simplifies to $\overrightarrow{AB} = (10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}$ km.

Show the boats do not collide

For collision, both components must be zero. From $10 - 24t = 0$, $t = \frac{5}{12}$. From $12t - 15 = 0$, $t = \frac{5}{4}$. These times are different, so the boats never collide.

Use the bearing 225 degrees condition

A bearing of 225° means B is south-west of A, so the $\mathbf{i}$ and $\mathbf{j}$ components of $\overrightarrow{AB}$ are equal and negative. Set $10 - 24t = 12t - 15$.

Find the distance

The equation gives $t = \frac{25}{36}$. Then each component is $-\frac{20}{3}$, so the distance is $\sqrt{\left(\frac{20}{3}\right)^2 + \left(\frac{20}{3}\right)^2} = \frac{20\sqrt{2}}{3}$ km.

Final answer (a) $\mathbf{v}_B = 16\mathbf{i} + 12\mathbf{j} \text{ km h}^{-1}$. (b) $\mathbf{s} = (10 + 16t)\mathbf{i} + (5 + 12t)\mathbf{j}$. (c) $\overrightarrow{AB} = (10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}$. (d) no collision. (e) $\frac{20\sqrt{2}}{3} \text{ km} \approx 9.4 \text{ km}$.

January 2025 Q5 — complete original question

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target.
The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

- (a) Find the velocity of the ball.

(3)

The position vector of the target is $(13\mathbf{i} - 5\mathbf{j})$ m.

- (b) Use the model to find the distance of B from the target at time $t = 7$ s.

(4)

DR ESLAM AHMED

January 2025 Q5 — detailed solution

Relative bearing and distance

ALL
7 marks**Find the displacement over 8 seconds**

From $t = 1$ to $t = 9$, the displacement is $(10\mathbf{i} - 3\mathbf{j}) - (-2\mathbf{i} + 5\mathbf{j}) = 12\mathbf{i} - 8\mathbf{j}$.

Find the velocity

The time interval is 8 seconds, so $\mathbf{v} = \frac{12\mathbf{i} - 8\mathbf{j}}{8} = 1.5\mathbf{i} - \mathbf{j} \text{ m s}^{-1}$.

Find the position at $t=7$

Time $t = 7$ is 6 seconds after $t = 1$. Therefore $\mathbf{r}_B(7) = (-2\mathbf{i} + 5\mathbf{j}) + 6(1.5\mathbf{i} - \mathbf{j}) = 7\mathbf{i} - \mathbf{j}$.

Find the vector from B to the target

The target has position vector $13\mathbf{i} - 5\mathbf{j}$. The vector from the ball to the target is $(13\mathbf{i} - 5\mathbf{j}) - (7\mathbf{i} - \mathbf{j}) = 6\mathbf{i} - 4\mathbf{j}$.

Find the distance

The distance is $\sqrt{6^2 + (-4)^2} = \sqrt{52} = 2\sqrt{13} \text{ m}$, which is about 7.2 m.

Final answer (a) $\mathbf{v} = 1.5\mathbf{i} - \mathbf{j} \text{ m s}^{-1}$. (b) distance = $2\sqrt{13} \text{ m} \approx 7.2 \text{ m}$.

Summer 2025 Q2 — complete original question

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of A . (2)

(b) Find an expression for $\mathbf{r}$ at time t seconds.
Give your answer in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are functions of t (2)

At time t seconds, the position vector of B is given by $\mathbf{s}$ metres where

$$\mathbf{s} = -5\mathbf{i} + 7\mathbf{j} + t(2\mathbf{i} - 3\mathbf{j})$$

(c) Find, to the nearest degree, the bearing of B from A when $t = 5$ (3)

Summer 2025 Q2 — detailed solution

Relative bearing and distance

ALL
7 marks**Part (a): Find velocity of A - Find change in position**

Given positions for particle A:

; At $t = 2$: $\mathbf{r} = -5\mathbf{i} + 16\mathbf{j}$; At $t = 5$: $\mathbf{r} = 10\mathbf{i} + 4\mathbf{j}$

$$\mathbf{v} = \frac{\Delta \mathbf{r}}{\Delta t}$$

$$\mathbf{v}_A = \frac{(10\mathbf{i} + 4\mathbf{j}) - (-5\mathbf{i} + 16\mathbf{j})}{5 - 2} = \frac{(10 + 5)\mathbf{i} + (4 - 16)\mathbf{j}}{3} = \frac{15\mathbf{i} - 12\mathbf{j}}{3} = 5\mathbf{i} - 4\mathbf{j} \text{ m s}^{-1}$$

Part (b): Find expression for $\mathbf{r}$ at time t - Use position-velocity relationship

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \mathbf{v}(t - t_0)$$

Using $t_0 = 2$ with $\mathbf{r}(2) = -5\mathbf{i} + 16\mathbf{j}$:

$$\begin{aligned}\mathbf{r}(t) &= (-5\mathbf{i} + 16\mathbf{j}) + (5\mathbf{i} - 4\mathbf{j})(t - 2) = -5\mathbf{i} + 16\mathbf{j} + 5(t - 2)\mathbf{i} - 4(t - 2)\mathbf{j} = -5\mathbf{i} + 16\mathbf{j} + (5t - 10)\mathbf{i} + (-4t + 8)\mathbf{j} \\ &= (-5 + 5t - 10)\mathbf{i} + (16 - 4t + 8)\mathbf{j} = (5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}\end{aligned}$$

Part (c): Find bearing of B from A when $t = 5$ - Find positions at $t = 5$ For A at $t = 5$: $\mathbf{r}_A = 10\mathbf{i} + 4\mathbf{j}$ (given)

$$\text{For B at } t = 5: \mathbf{s}_B = -5\mathbf{i} + 7\mathbf{j} + 5(2\mathbf{i} - 3\mathbf{j}) = -5\mathbf{i} + 7\mathbf{j} + 10\mathbf{i} - 15\mathbf{j} = 5\mathbf{i} - 8\mathbf{j}$$

Part (c): Find bearing of B from A when $t = 5$ - Find vector from A to B

$$\overrightarrow{AB} = \mathbf{s}_B - \mathbf{r}_A = (5\mathbf{i} - 8\mathbf{j}) - (10\mathbf{i} + 4\mathbf{j}) = -5\mathbf{i} - 12\mathbf{j}$$

Part (c): Find bearing of B from A when $t = 5$ - Find angle and bearingAngle from South (negative $\mathbf{j}$ direction): $\tan \alpha = \frac{5}{12}$ $\alpha = \tan^{-1}\left(\frac{5}{12}\right)$ $\alpha = 22.619\dots^\circ$

$$\text{Bearing} = 180^\circ + \alpha = 180^\circ + 22.619\dots^\circ = 202.619\dots^\circ$$

Final answer (a) $\mathbf{v}_A = 5\mathbf{i} - 4\mathbf{j} \text{ m s}^{-1}$. (b) $\mathbf{r} = (5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}$. (c) bearing = 203° .

January 2026 Q6 — complete original question

ELITE EDITORIAL CORRECTION — SOURCE-BACKED

In part (e), read “south-west” instead of the printed “north-east”.

Why: the final mark scheme flags an SW/NE error. Equating the i - and j -components gives $t = 40/13$ and $\vec{SB} = -(70/13)\mathbf{i} - (70/13)\mathbf{j}$, so B is south-west of S .

The original Pearson question pixels are preserved unchanged below.

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 9 am, a ship S leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j})\text{ km h}^{-1}$

- (a) Find the exact speed of S (2)

The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j})\text{ km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j})\text{ km h}^{-1}$

- (b) Find the direction in which B is moving, giving your answer as a bearing to the nearest degree. (3)

- (c) Show that at time t hours after 9 am,

$$\vec{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}]\text{ km} \quad (4)$$

- (d) Show that B and S do not collide. (3)

When B is north-east of S , the distance between S and B is $d\text{ km}$.

- (e) Find the value of d . (4)

January 2026 Q6 — detailed solution

Relative bearing and distance

ALL
16 marks**Find the speed of S**

The velocity of S is $10\mathbf{i} + 15\mathbf{j}$. Its speed is the magnitude: $\sqrt{10^2 + 15^2} = \sqrt{325} = 5\sqrt{13} \text{ km h}^{-1}$.

Find the bearing of B

B has velocity $-8\mathbf{i} + 10\mathbf{j}$, so it moves west and north. The angle west of north satisfies $\tan \theta = \frac{8}{10}$, giving $\theta = 38.66^\circ$. The bearing is $360^\circ - 38.66^\circ = 321^\circ$ to the nearest degree.

Write both position vectors

At time t hours after 9 am, $\mathbf{r}_S = (10t)\mathbf{i} + (15t)\mathbf{j}$. Also $\mathbf{r}_B = (50\mathbf{i} + 10\mathbf{j}) + t(-8\mathbf{i} + 10\mathbf{j}) = (50 - 8t)\mathbf{i} + (10 + 10t)\mathbf{j}$.

Find the relative position vector

$\vec{SB} = \mathbf{r}_B - \mathbf{r}_S = [(50 - 8t) - 10t]\mathbf{i} + [(10 + 10t) - 15t]\mathbf{j} = (50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}$.

Show they do not collide

For a collision, both components of $\vec{SB}$ must be zero at the same time. From $50 - 18t = 0$, $t = \frac{25}{9}$. From $10 - 5t = 0$, $t = 2$. These are different, so B and S do not collide.

Apply the source-backed south-west correction

The final mark scheme identifies the printed north-east wording as the SW/NE error. Use south-west, so the two relative-position components are equal and negative: $50 - 18t = 10 - 5t$. Hence $40 = 13t$ and $t = 40/13$.

Find the distance d

At $t = \frac{40}{13}$, both components are $-\frac{70}{13}$. Therefore $d = \sqrt{\left(\frac{70}{13}\right)^2 + \left(\frac{70}{13}\right)^2} = \frac{70\sqrt{2}}{13} = 7.6 \text{ km}$ to 2 significant figures.

Final answer (a) $5\sqrt{13} \text{ km h}^{-1}$. (b) 321° . (c) $\vec{SB} = (50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}$. (d) no collision. (e) $d = \frac{70\sqrt{2}}{13} \text{ km} \approx 7.6 \text{ km}$.

Summer 2026 Q7 — complete original question

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At time $t = 1.5$ hours after noon, the boat passes through the point with position vector $(11\mathbf{i} + 3.5\mathbf{j}) \text{ km}$.

- (a) Show that, at time t hours after noon, the position vector of the boat is

$$[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}] \text{ km} \quad (3)$$

The boat is visible from O when it is 9 km or less from O .

- (b) Find the length of time for which the boat is visible from O , showing your method clearly. (4)
- (c) Find the value of t when the boat is northwest of O . (2)

At the instant when the boat is at the point P , OP is perpendicular to the direction in which the boat is moving.

- (d) Show that $OP = \frac{18\sqrt{5}}{5} \text{ km}$ (6)
- (Solutions relying on calculator technology are not acceptable.)

Summer 2026 Q7 — detailed solution

Relative bearing and distance

ALL
15 marks**Part (a): build the position vector**

From the known position at $t = 1.5$, $\mathbf{r} = (11\mathbf{i} + 3.5\mathbf{j}) + (t - 1.5)(-2\mathbf{i} + \mathbf{j}) = [(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}]$ km.

Part (b): enter and leave the visibility circle

At the two boundary times, $(14 - 2t)^2 + (2 + t)^2 = 81$, so $5t^2 - 52t + 119 = 0$. The roots are 3.4 and 7, hence the visible duration is 3.6 h.

Part (c): translate northwest

Northwest means a negative $\mathbf{i}$ component and an equal positive $\mathbf{j}$ component. Thus $14 - 2t = -(2 + t)$, giving $t = 16$.

Part (d): closest-point perpendicularity

At closest approach, $\mathbf{r} \cdot (-2\mathbf{i} + \mathbf{j}) = 0$. Hence $-2(14 - 2t) + (2 + t) = 0$, so $t = 26/5$ and $\overrightarrow{OP} = (18/5)\mathbf{i} + (36/5)\mathbf{j}$.

Evaluate the exact distance

Therefore $OP = \sqrt{(18/5)^2 + (36/5)^2} = 18\sqrt{5}/5$ km.

Final answer (a) $[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}]$ km. (b) 3.6 h. (c) $t = 16$. (d) $OP = 18\sqrt{5}/5$ km.

January 2021 Q6 — complete original question

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{m s}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{m s}^{-1}$

- (a) Find the position vector of Agatha at time t seconds.

(2)

At time $t = 6$ seconds, Agatha passes through the point P .

- (b) Show that Brionie also passes through P and find the value of t when this occurs.

(4)

At time t seconds, Agatha is at the point A and Brionie is at the point B .

- (c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$

(2)

- (d) Find the distance between the two girls when they are closest together.

(4)

January 2021 Q6 — detailed solution

Infer a landmark or route

ALL
12 marks**Write Agatha position**

$$\mathbf{r}_A = (11\mathbf{i} + 11\mathbf{j}) + t(3\mathbf{i} - \mathbf{j}) = (11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}.$$

Find point P

At $t = 6$, Agatha is at $(11 + 18)\mathbf{i} + (11 - 6)\mathbf{j} = 29\mathbf{i} + 5\mathbf{j}$.

Write Brionie position

$$\mathbf{r}_B = (7\mathbf{i} + 16\mathbf{j}) + t(4\mathbf{i} - 2\mathbf{j}) = (7 + 4t)\mathbf{i} + (16 - 2t)\mathbf{j}.$$

Show Brionie passes P

Set $7 + 4t = 29$ to get $t = 5.5$. Also $16 - 2t = 5$ gives $t = 5.5$. So Brionie also passes through P at $t = 5.5$ s.

Find AB vector

$$\overrightarrow{AB} = \mathbf{r}_B - \mathbf{r}_A = [7 + 4t - (11 + 3t)]\mathbf{i} + [16 - 2t - (11 - t)]\mathbf{j}.$$

Simplify AB

So $\overrightarrow{AB} = (t - 4)\mathbf{i} + (5 - t)\mathbf{j}$ m.

Minimise distance squared

$$AB^2 = (t - 4)^2 + (5 - t)^2 = 2(t - 4.5)^2 + 0.5.$$

State the minimum distance

The minimum occurs when $t = 4.5$, giving $AB^2 = 0.5$. Hence the minimum distance is $\sqrt{0.5} = 0.707 \dots$ m.

Final answer (a) $\mathbf{r}_A = (11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}$. (b) $t = 5.5$ s. (c) $\overrightarrow{AB} = (t - 4)\mathbf{i} + (5 - t)\mathbf{j}$. (d) minimum distance = $\sqrt{0.5}$ m ≈ 0.71 m.

October 2021 Q8 — complete original question

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

- (a) Show that the velocity of the ship is $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$ (2)

- (b) Find the position vector of the ship t hours after leaving port. (2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse. (4)

- (d) Find the position vector of the ship when it is due south of the lighthouse. (4)

(Total 12 marks)

October 2021 Q8 — detailed solution

Infer a landmark or route

ALL
12 marks**Convert the time**

From 7:00 to 7:36 is 36 minutes, which is 0.6 hours.

Find velocityDisplacement = $(4\mathbf{i} + 6\mathbf{j}) - (-2\mathbf{i} + 9\mathbf{j}) = 6\mathbf{i} - 3\mathbf{j}$. Divide by 0.6: $\mathbf{v} = 10\mathbf{i} - 5\mathbf{j}$.**Write the ship position** $\mathbf{r} = (-2\mathbf{i} + 9\mathbf{j}) + t(10\mathbf{i} - 5\mathbf{j}) = (-2 + 10t)\mathbf{i} + (9 - 5t)\mathbf{j}$.**Use 8:48**At 8:48, $t = 1.8$. The ship position is $(-2 + 18)\mathbf{i} + (9 - 9)\mathbf{j} = 16\mathbf{i}$. The lighthouse is due east, so it has the same j-component, 0.**Use 9:00**At 9:00, $t = 2$. The ship position is $18\mathbf{i} - \mathbf{j}$. The lighthouse is north-east of the ship, so its east and north displacements from the ship are equal.**Find the lighthouse position**Since the lighthouse has j-component 0, from $18\mathbf{i} - \mathbf{j}$ it is 1 km north, so it is also 1 km east. Hence its position is $19\mathbf{i}$.**Find when the ship is due south**Due south of the lighthouse means the ship has i-component 19. Set $-2 + 10t = 19$, so $t = 2.1$.**Find the ship position**At $t = 2.1$, $\mathbf{r} = (-2 + 21)\mathbf{i} + (9 - 10.5)\mathbf{j} = 19\mathbf{i} - 1.5\mathbf{j}$.**Final answer** (a) $\mathbf{v} = 10\mathbf{i} - 5\mathbf{j} \text{ km h}^{-1}$. (b) $\mathbf{r} = (-2 + 10t)\mathbf{i} + (9 - 5t)\mathbf{j}$. (c) lighthouse = $19\mathbf{i}$. (d) ship position = $19\mathbf{i} - 1.5\mathbf{j}$.

October 2021 Q4 — complete original question

4. The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O , is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

- (a) Find, to the nearest degree, the size of the angle between $\mathbf{r}$ and the vector $\mathbf{j}$, when $t = 2$
(3)

- (b) Find the values of t for which the distance of P from O is 2.5 m.
(5)

DR ESLAM AHMED

October 2021 Q4 — detailed solution

Position-vector distance locus

ALL
8 marks**Find $\mathbf{r}$ when $t = 2$**

$$\mathbf{r} = (2 - 3)\mathbf{i} + (1 - 2(2))\mathbf{j} = -\mathbf{i} - 3\mathbf{j}.$$

Find the angle with $\mathbf{j}$

The vector is in the third quadrant. The angle between $\mathbf{r}$ and the positive $\mathbf{j}$ direction satisfies $\tan \alpha = \frac{1}{3}$ from the vertical comparison.

State the angle

$\alpha = 18.4^\circ$ from the negative $\mathbf{j}$ direction, so the size of the angle with positive $\mathbf{j}$ is $180^\circ - 18.4^\circ = 161.6^\circ$, which is 162° .

Use the distance condition

Distance from O is 2.5, so $(t - 3)^2 + (1 - 2t)^2 = 2.5^2$.

Form the quadratic

$$t^2 - 6t + 9 + 1 - 4t + 4t^2 = 6.25, \text{ so } 5t^2 - 10t + 3.75 = 0.$$

Solve

Multiply by 4: $20t^2 - 40t + 15 = 0$, or $4t^2 - 8t + 3 = 0$. Hence $(2t - 1)(2t - 3) = 0$, so $t = \frac{1}{2}$ or $t = \frac{3}{2}$.

Final answer (a) 162° . (b) $t = \frac{1}{2}$ or $t = \frac{3}{2}$.

TOPIC 02 | FAMILY 04

Closest approach in relative motion

WHAT THIS FAMILY ASKS

Minimise the separation of particles moving at constant vector velocities.

VARIANT ROUTE

V01 Minimise distance squared

1 classified questions

11 total marks

January 2024 Q7 — complete original question

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At midnight, a ship S is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})$ km

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j}) \text{ km h}^{-1}$

- (a) Find the speed of S .

(2)

At time t hours after midnight, the position vector of S is $\mathbf{s}$ km.

- (b) Find an expression for $\mathbf{s}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector $(26\mathbf{i} + 15\mathbf{j})$ km.

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of t when S is closest to the lighthouse.

- (ii) Hence determine whether it is safe for S to continue its course.

(7)

January 2024 Q7 — detailed solution

Minimise squared distance

ALL
11 marks**Find the speed**

The velocity is $12\mathbf{i} - 16\mathbf{j}$, so the speed is $\sqrt{12^2 + (-16)^2} = 20 \text{ km h}^{-1}$.

Write the ship position vector

At time t hours after midnight, $\mathbf{s} = (19\mathbf{i} + 22\mathbf{j}) + t(12\mathbf{i} - 16\mathbf{j}) = (19 + 12t)\mathbf{i} + (22 - 16t)\mathbf{j}$.

Form the vector from lighthouse to ship

The lighthouse is at $26\mathbf{i} + 15\mathbf{j}$. Therefore $\overrightarrow{LS} = \mathbf{s} - \mathbf{l} = (12t - 7)\mathbf{i} + (7 - 16t)\mathbf{j}$.

Use distance squared

The square of the distance is $d^2 = (12t - 7)^2 + (7 - 16t)^2 = 400t^2 - 392t + 98$.

Find the time of closest approach

This quadratic is minimum when $t = \frac{392}{2 \cdot 400} = 0.49$.

Find the minimum distance

At $t = 0.49$, $d^2 = 1.96$, so $d = 1.4 \text{ km}$.

Decide if the course is safe

The unsafe distance is 1.3 km . Since $1.4 > 1.3$, it is safe for the ship to continue its course.

Final answer (a) 20 km h^{-1} . (b) $\mathbf{s} = (19 + 12t)\mathbf{i} + (22 - 16t)\mathbf{j}$. (c)(i) $t = 0.49$. (c)(ii) minimum distance = 1.4 km , so it is safe.